

Female Reservation and the Qualifications of Local Elected Officials in India

Gaurav Sood

Working draft

1 Introduction

Among winners with classifiable education, female reservation is associated with lower graduate shares among village heads in Bihar, Uttar Pradesh and Rajasthan. Rajasthan’s missing education is too extensive to determine the direction of the all-winner difference. Kerala’s village ward members show much smaller differences. Urban evidence points in another direction: Mumbai’s reserved-seat councillors have a higher estimated graduate share and fewer pending criminal cases. This paper studies the effect of reserving an office for women on the qualifications of the person elected to it. Rural panchayats and urban councils are analyzed separately.

The graduate-share difference for Bihar’s mukhiyas is -15.0 percentage points (95% CI -16.6 to -13.5). The corresponding estimates in UP are negative in each election analyzed. Kerala’s ward-member estimates are close to zero, with intervals that include both gains and losses. Mumbai’s graduate-share interval also includes zero. These comparisons describe specific offices and elections; they do not establish a single national effect.

Education, political experience and pending criminal cases measure different attributes of elected officials. Age describes their composition without ranking competence. The analysis retains these distinctions rather than combining them into a quality index. It also keeps leader qualifications separate from performance in office. A reservation can change both, but a result for public spending or corruption does not establish how reservation changes entrants’ qualifications.

2 Estimand and evidence

The target is the impact of female reservation on the elected officeholder’s qualification, relative to opening the same office to either sex. An open seat can elect a woman. Comparing female and male winners would answer a different question, as would comparing all candidates rather than the people elected.

For each rural state, election and office, the analysis estimates

$$Y_i = \alpha + \beta R_i + \gamma_{g(i)} + \delta_{c(i)} + \epsilon_i,$$

where R_i records female seat reservation, $\gamma_{g(i)}$ controls for geography and $\delta_{c(i)}$ for caste reservation. Geography is the block when it is populated throughout the relevant slice, and otherwise the district.

The district and block together identify a block. Standard errors are clustered at the geographic-control level. All reported intervals are pointwise 95% intervals from `fixest`, using its default small-sample adjustment. They do not adjust for testing several offices or outcomes.

The coefficient β is an adjusted reserved-versus-open difference. Its causal interpretation requires conditional comparability of reservation assignments and adequate outcome coverage. Geographic and caste controls alone do not verify that requirement. No lottery or discontinuity design is claimed for these administrative-data regressions. Mumbai uses administrative-ward and council effects, with standard errors clustered by electoral ward.

The administrative inputs flow from state election repositories through `local_reservations`. The analysis records input hashes, winner selection, exclusions, outcome classification and regression samples. Bihar’s winners are inferred from unique highest recorded votes or sole uncontested candidacies. UP’s earlier files report elected pradhans; its latest candidate file marks winners. Rajasthan links nomination characteristics to results. Kerala records elected ward members, including those who subsequently hold council leadership positions. The reservation variable for those records refers to their ward, not the presidency.

3 Rural panchayats

3.1 Village heads

Rural village heads: graduate or above

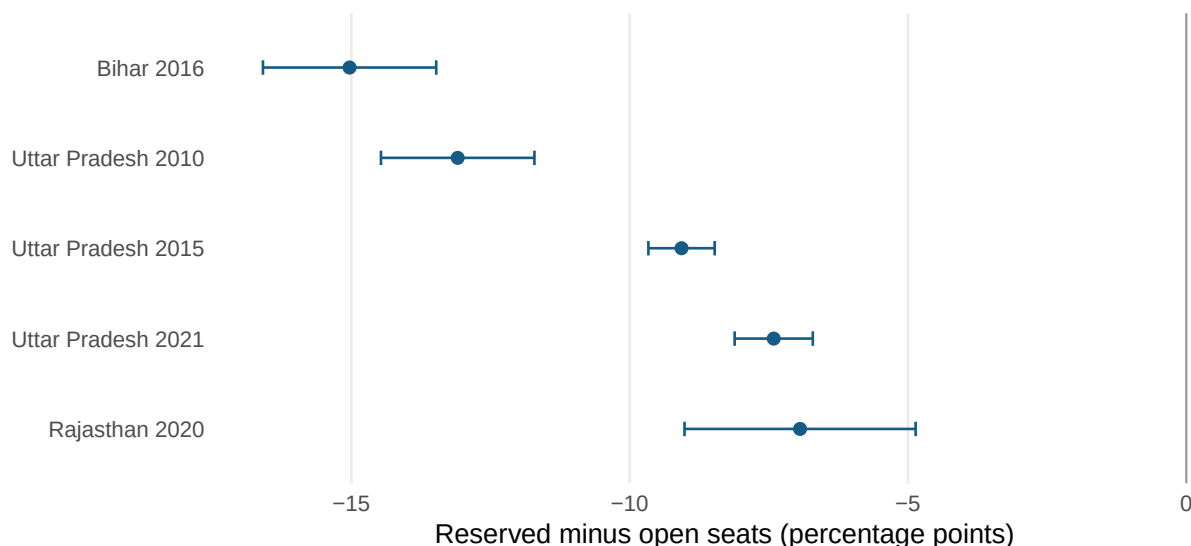


Figure 1: Graduate-share differences among rural village heads. Points compare women-reserved with open seats, conditional on geography and caste reservation; bars are pointwise 95% confidence intervals clustered by geography. UP 2010 uses districts; other rows use blocks. Education is restricted to classified qualifications. Ambiguous winners and unknown reservation are excluded. Sources: state election records through the central election repository.

Table 1: Rural village heads: graduate-share differences in percentage points.

Election	Difference	95% CI	N
Bihar 2016	-15.0	[-16.6, -13.5]	7,908
Uttar Pradesh 2010	-13.1	[-14.5, -11.7]	51,566
Uttar Pradesh 2015	-9.1	[-9.7, -8.5]	59,019
Uttar Pradesh 2021	-7.4	[-8.1, -6.7]	47,228
Rajasthan 2020	-6.9	[-9.0, -4.9]	7,748

UP’s graduate-share gap is -13.1 percentage points (95% CI -14.5 to -11.7) in 2010, -9.1 percentage points (95% CI -9.7 to -8.5) in 2015, and -7.4 percentage points (95% CI -8.1 to -6.7) in 2021. These are separate election samples. Changes in their coefficients do not by themselves identify learning, convergence or an effect of repeated reservation.

Rajasthan’s estimate is -6.9 percentage points (95% CI -9.0 to -4.9). Its source includes many qualifications labelled “Other”, which remain unclassified. The estimate therefore describes winners whose graduate status can be classified. Allowing every missing qualification to take either degree status puts the adjusted difference for all eligible winners between -29.7 and 31.5 percentage points. These data alone do not determine its direction.

3.2 Bihar’s other offices

Bihar provides evidence across village, block and district offices. Its analysis includes 210,910 elected seats after excluding ambiguous candidate serials. Unknown education remains missing. Implausible or missing ages do not remove a winner from education regressions.

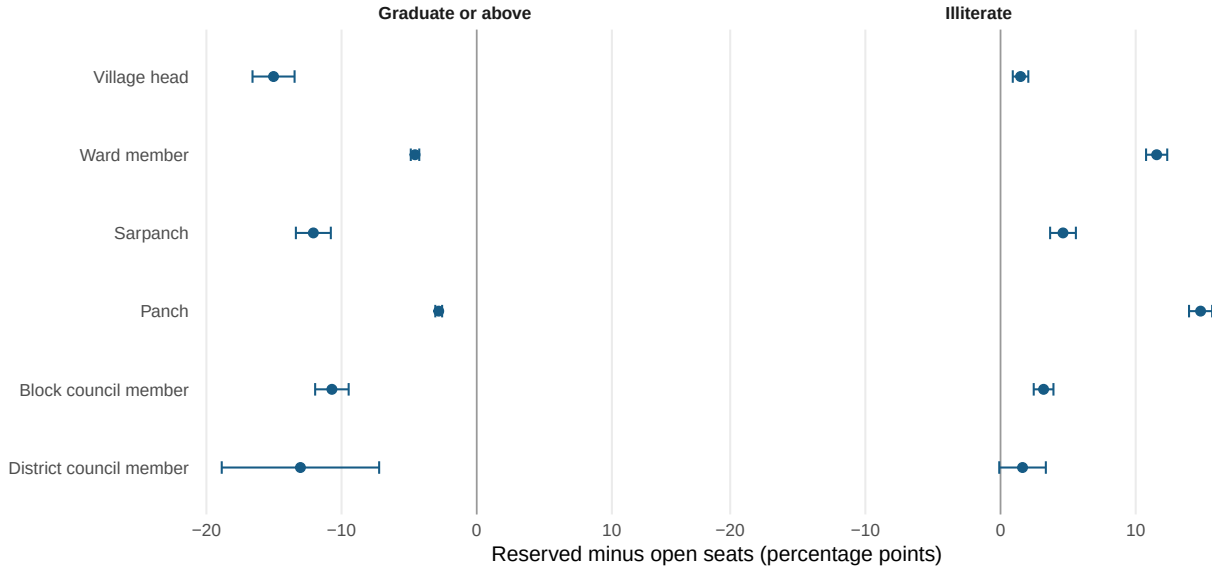
Bihar 2016: elected officials

Figure 2: Bihar 2016: education of elected officials by office. Points are reserved-minus-open differences with block and caste-reservation controls, except district council members, who use districts. Bars are pointwise 95% confidence intervals clustered at that geographic level. Unknown education and ambiguous winner records are excluded. Source: Bihar SEC candidate records through the central election repository.

The graduate-share estimate is negative in every office. Illiteracy is higher in reserved seats, although the district-council estimate is imprecise. The separate age regressions describe younger winners under reservation; age is not treated as an ordinal measure of quality.

3.3 Kerala by election and office

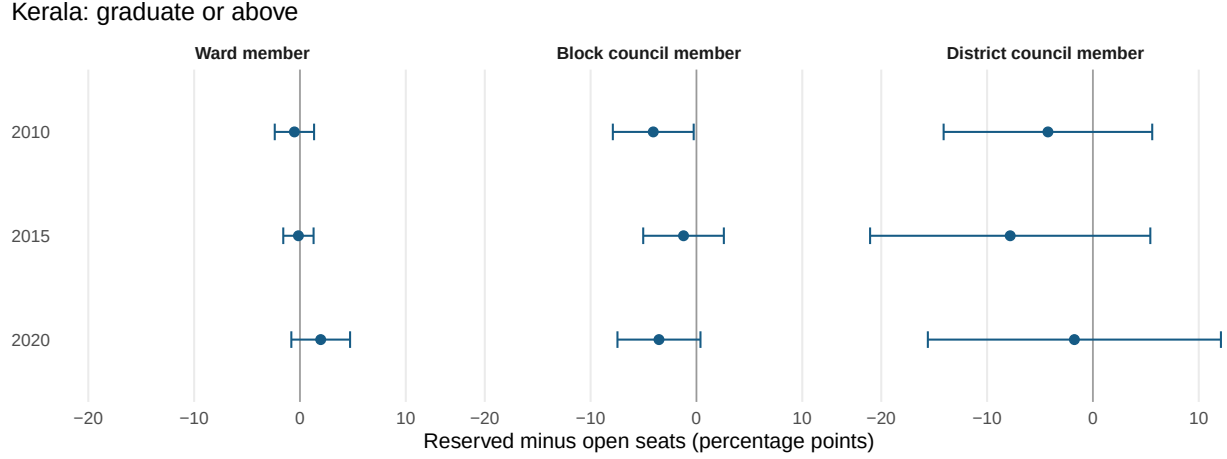


Figure 3: Rural Kerala: graduate-share differences by election and office. Each point comes from a separate regression with caste-reservation controls. Block council members use block effects and clustering; village and district ward members use districts because block identifiers are unavailable. Bars are pointwise 95% confidence intervals. Unclassified qualifications are missing. The common horizontal scale permits comparison across offices. Source: Kerala election records through the central election repository.

For Kerala’s village ward members, the differences are -0.5 percentage points (95% CI -2.4 to 1.3), -0.1 percentage points (95% CI -1.6 to 1.3), and 2.0 percentage points (95% CI -0.8 to 4.7). They provide no precise evidence of the large negative gaps observed among village heads elsewhere. They concern ward members, however, so the contrast with Bihar and UP cannot be attributed to the state alone.

Education is recorded as free text. The coding recognizes reported degrees and school qualifications, leaves incomplete or unclassified qualifications missing, and exports every source label with its classification and frequency. It does not infer that an unmatched string means no degree. These rules differ from the earlier pooled Kerala descriptive table, which used loose substring matches and included unmatched values in its denominator. That historical table is preserved in the repository’s evidence archive rather than treated as an estimate from this specification.

4 Urban councils

4.1 Mumbai

Mumbai’s data link reservation to councillor qualifications in the Praja/affidavit records held in the Karekurve-Ramachandra and Lee replication deposit (Karekurve-Ramachandra and Lee 2023). The sample contains one observation per councillor spell with age recorded. Repeated survey waves do not create additional councillors. The analysis pools the 2012 and 2017 councils with council and administrative-ward effects; standard errors are clustered by electoral ward.

The graduate-share difference is 8.4 percentage points (95% CI -0.4 to 17.3). The pending-case

Urban: Mumbai councillors, 2012 and 2017

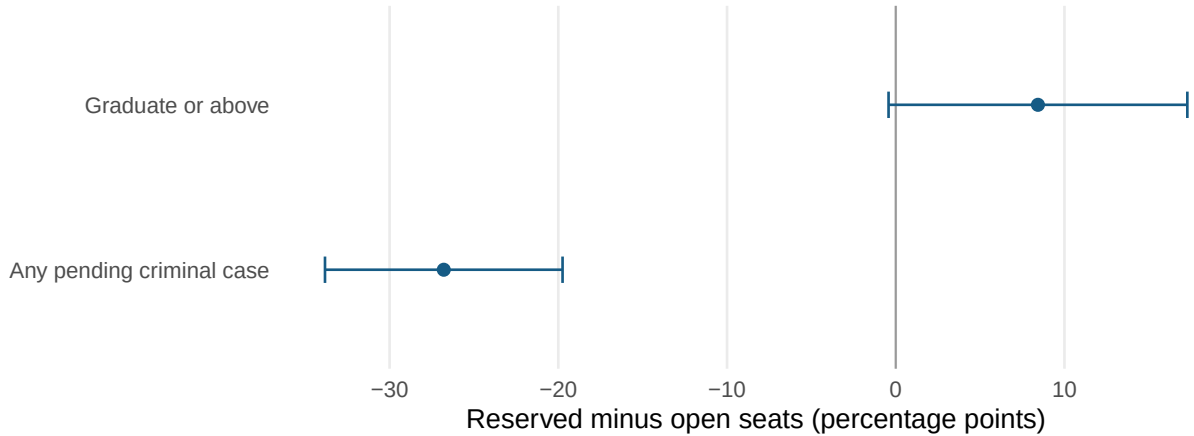


Figure 4: Urban Mumbai: qualifications of elected councillors in the 2012 and 2017 councils. Points are women-reserved-minus-open differences with administrative-ward and council effects; bars are pointwise 95% confidence intervals clustered by electoral ward. One observation is retained per councillor spell with age recorded. Pending cases are allegations, not convictions. Source: Praja/affidavit data in the Karekurve-Ramachandra and Lee replication deposit.

difference is -26.8 percentage points (95% CI -33.8 to -19.7). The education interval permits a small negative effect as well as a positive one; it does not establish that reservation raises education. Pending criminal cases are separately reported because a charge does not establish guilt.

The earlier Rajasthan municipal comparison and Kerala’s urban descriptive tables belong to this urban evidence, not the rural estimates above. Rajasthan’s category “high school or less” includes the twelfth standard; Mumbai’s ends at the tenth. Those levels cannot be pooled as a common outcome. The historical tables remain available in the evidence archive, with their original denominators and qualifications.

5 What the existing studies establish

Lower schooling and less experience under reservation appear in several rural studies. In the 2001 working paper’s West Bengal sample, reserved council heads have 7.13 rather than 9.92 years of schooling, a difference of -2.79 years (SE 0.54); 11% rather than 43% had previously been elected to the council (Chattopadhyay and Duflo 2001, Table 3). In the subsequent Birbhum council term, the adjusted schooling gap is -2.10 years (SE 0.55), and reserved leaders are 26.5 percentage points more likely to be first-time pradhans (SE 6.8) (Beaman et al. 2009, Table II, column 6). These samples overlap geographically and are not independent national tests.

The May 2007 southern-state draft, published in 2008, reports -2.62 years of schooling (SE 0.77) and a political-knowledge gap of -1.41 points (SE 0.29). That regression excludes caste-reserved councils (Ban and Rao 2007, Table 9). Karnataka’s descriptive comparison reports degrees among none of the 23 reserved female presidents and among 19% of the 48 male presidents. Illiteracy is 9% and 4%, respectively; the comparison also conditions on the leader’s gender, so it is not a clean reservation contrast (Raabe et al. 2009, Tables 3–4).

The 2011 working paper on the NCAER village panel, published in 2015, reports illiteracy of 24.72%

in reserved observations versus 5.74% in unreserved observations, secondary education or above of 28.65% versus 62.69%, and previous officeholding of 11.67% versus 24.24%. These are descriptive comparisons across 180 reserved and 459 unreserved observations (Deininger et al. 2011, Table 3). The 2013 Andhra Pradesh working paper, published in 2017, reports that 15% of reserved heads versus 54% of unreserved heads have higher-secondary schooling or more; 11% versus 22% have prior political experience. The characteristics table covers 128 reserved and 169 unreserved heads (Afridi et al. 2013, Table 2).

Recent studies require additional distinctions. The September 2022 Karnataka candidate draft reports lower absolute qualification distributions under reservation (Mori et al. 2022, fig. 2 and Table 3). The published article reports stronger cognitive selection relative to the eligible female population (Mori et al. 2025). That relative benchmark does not establish that reservation increases the elected elite’s absolute qualifications. It also concerns candidates. The March 2026 Bihar threshold draft reports standardized schooling effects of -0.765 SD for candidates (SE 0.024; $N=44,583$) and -0.921 SD for winners (SE 0.073; $N=3,294$). Substituting husbands’ schooling gives -0.147 SD and -0.364 SD, respectively (Bamezai et al. 2026, Table 7). Spousal schooling does not reveal who governs.

Favorable judgments about participation, voter attitudes or policy performance address different outcomes. They can coexist with lower schooling among entrants. Likewise, a positive selection score relative to same-gender citizens can coexist with lower absolute schooling. A synthesis of the reservation effect must keep the outcome, elected status, comparison group and office fixed before combining estimates. The studies reviewed here do not yet supply a harmonized meta-analysis.

6 Measurement and reproducibility

Table 2: Education coverage among eligible winner records. N includes unclassified qualifications.

State	Year	Office	Seat	Classified (%)	N
UP	2010	Village head	Open	99.6	34,654
UP	2010	Village head	Women	99.4	17,166
UP	2015	Village head	Open	100.0	39,093
UP	2015	Village head	Women	100.0	19,926
UP	2021	Village head	Open	95.3	32,997
UP	2021	Village head	Women	94.1	16,774
Rajasthan	2020	Village head	Open	81.8	5,695
Rajasthan	2020	Village head	Women	56.9	5,433
Kerala	2010	Block council member	Open	91.0	983
Kerala	2010	Block council member	Women	93.0	1,095
Kerala	2010	Ward member	Open	89.7	8,047
Kerala	2010	Ward member	Women	90.8	8,594
Kerala	2010	District council member	Open	87.7	162
Kerala	2010	District council member	Women	94.7	169
Kerala	2015	Block council member	Open	87.4	1,021
Kerala	2015	Block council member	Women	88.7	1,041
Kerala	2015	Ward member	Open	87.6	7,725
Kerala	2015	Ward member	Women	89.7	8,196
Kerala	2015	District council member	Open	86.0	164

State	Year	Office	Seat	Classified (%)	N
Kerala	2015	District council member	Women	92.2	167
Kerala	2020	Block council member	Open	88.7	980
Kerala	2020	Block council member	Women	89.3	1,098
Kerala	2020	Ward member	Open	83.8	7,726
Kerala	2020	Ward member	Women	84.9	8,226
Kerala	2020	District council member	Open	86.1	166
Kerala	2020	District council member	Women	84.2	165

The confidence intervals reflect regression uncertainty conditional on the extracted and classified data. They do not include transcription error, uncertainty about free-text qualifications, or selection from missing education. Rajasthan’s unspecified qualifications and Kerala’s differential classification rates limit claims about all elected officials. Bihar’s age range is restricted to 21–100, as are the new rural age analyses; the education samples retain those with missing age.

The scripts retain source identifiers, sample counts, recoding dictionaries and typed Parquet analysis files. Input hashes pin the local files used here. Some UP inputs come from integration work whose central adapter has uncommitted changes; the repository preserves the adapter diff and byte hashes rather than presenting that state as a published release. The final data release should reconcile those changes upstream. The included analysis snapshots allow inspection of the fitted samples in the meantime.

The 2021 Bihar candidate release lacks statewide structured winner education. Its current reservation feeds and historical source checks are retained upstream; no 2021 Bihar reservation-effect estimate enters this paper. Delhi’s municipal files are a separate potential urban extension whose source provenance and field coding still need verification.

The missing-education bounds hold the all-winner regression design fixed and allow each unclassified graduate indicator to be either zero or one. After residualizing reservation on the controls, the coefficient is a weighted sum of outcomes. Assigning missing outcomes according to the signs of those weights gives the exact minimum and maximum coefficient. These bounds concern the adjusted coefficient, not the causal effect, and contain no sampling uncertainty. UP’s bounds remain negative in each election; Rajasthan’s and Kerala’s include both signs.

Table 3: Missing-education bounds on the adjusted all-winner graduate-share difference (percentage points). These are not confidence intervals.

State	Year	Office	Lower	Upper	Missing	N
UP	2010	Village head	-13.5	-12.4	254	51,820
UP	2015	Village head	-9.1	-9.1	0	59,019
UP	2021	Village head	-12.0	-1.3	2,543	49,771
Rajasthan	2020	Village head	-29.7	31.5	3,380	11,128
Kerala	2010	Block council member	-12.3	3.4	165	2,078
Kerala	2010	Ward member	-10.6	8.9	1,617	16,641
Kerala	2010	District council member	-12.7	4.8	29	331
Kerala	2015	Block council member	-13.8	10.1	247	2,062
Kerala	2015	Ward member	-12.2	10.5	1,800	15,921
Kerala	2015	District council member	-18.0	3.7	36	331

State	Year	Office	Lower	Upper	Missing	N
Kerala	2020	Block council member	-14.5	7.5	228	2,078
Kerala	2020	Ward member	-14.4	16.9	2,495	15,952
Kerala	2020	District council member	-15.1	15.2	49	331

7 Appendix: complete rural estimates

7.1 Bihar

Table 4: Bihar rural officials. Graduate and illiterate: percentage points; age: years. 95% intervals clustered by geography; caste controls included.

Year	Office	Outcome	Difference	95% CI	N	Controls
2016	Village head	Graduate+	-15.0	[-16.6, -13.5]	7,908	Block
2016	Village head	Illiterate	1.5	[0.9, 2.1]	7,908	Block
2016	Village head	Age	-3.6	[-4.1, -3.1]	7,969	Block
2016	Panch	Graduate+	-2.8	[-3.1, -2.6]	85,385	Block
2016	Panch	Illiterate	14.8	[13.9, 15.7]	85,385	Block
2016	Panch	Age	-2.9	[-3.1, -2.7]	85,406	Block
2016	Block member	Graduate+	-10.7	[-12.0, -9.5]	10,769	Block
2016	Block member	Illiterate	3.2	[2.5, 3.9]	10,769	Block
2016	Block member	Age	-2.7	[-3.1, -2.3]	10,801	Block
2016	Sarpanch	Graduate+	-12.1	[-13.4, -10.8]	7,864	Block
2016	Sarpanch	Illiterate	4.6	[3.7, 5.6]	7,864	Block
2016	Sarpanch	Age	-5.7	[-6.2, -5.2]	7,894	Block
2016	Ward member	Graduate+	-4.6	[-4.9, -4.3]	97,278	Block
2016	Ward member	Illiterate	11.6	[10.8, 12.3]	97,278	Block
2016	Ward member	Age	-2.0	[-2.2, -1.8]	97,289	Block
2016	District member	Graduate+	-13.0	[-18.9, -7.2]	889	District
2016	District member	Illiterate	1.6	[-0.1, 3.4]	889	District
2016	District member	Age	-4.1	[-5.6, -2.6]	892	District

7.2 Uttar Pradesh

Table 5: Uttar Pradesh rural officials. Graduate and illiterate: percentage points; age: years. 95% intervals clustered by geography; caste controls included.

Year	Office	Outcome	Difference	95% CI	N	Controls
2010	Village head	Graduate+	-13.1	[-14.5, -11.7]	51,566	District
2010	Village head	Illiterate	6.3	[5.1, 7.4]	51,566	District
2010	Village head	Age	-1.4	[-1.6, -1.2]	51,662	District
2015	Village head	Graduate+	-9.1	[-9.7, -8.5]	59,019	Block
2015	Village head	Illiterate	9.3	[8.7, 10.0]	59,019	Block
2021	Village head	Graduate+	-7.4	[-8.1, -6.7]	47,228	Block
2021	Village head	Illiterate	8.2	[7.5, 9.0]	47,228	Block

Year	Office	Outcome	Difference	95% CI	N	Controls
2021	Village head	Age	-1.3	[-1.5, -1.1]	49,757	Block

7.3 Rajasthan

Table 6: Rajasthan rural officials. Graduate and illiterate: percentage points; age: years. 95% intervals clustered by geography; caste controls included.

Year	Office	Outcome	Difference	95% CI	N	Controls
2020	Village head	Graduate+	-6.9	[-9.0, -4.9]	7,748	Block
2020	Village head	Illiterate	0.1	[-0.1, 0.2]	7,748	Block
2020	Village head	Age	-1.7	[-2.3, -1.1]	11,113	Block

7.4 Kerala

Table 7: Kerala rural officials. Graduate and illiterate: percentage points; age: years. 95% intervals clustered by geography; caste controls included.

Year	Office	Outcome	Difference	95% CI	N	Controls
2010	Block member	Graduate+	-4.1	[-7.9, -0.3]	1,913	Block
2010	Block member	Age	-4.9	[-5.8, -4.0]	1,980	Block
2010	Ward member	Graduate+	-0.5	[-2.4, 1.3]	15,024	District
2010	Ward member	Age	-4.8	[-5.9, -3.8]	14,793	District
2010	District member	Graduate+	-4.3	[-14.1, 5.6]	302	District
2010	District member	Age	-4.2	[-7.3, -1.2]	330	District
2015	Block member	Graduate+	-1.2	[-5.0, 2.6]	1,815	Block
2015	Block member	Age	-3.9	[-4.9, -3.0]	2,029	Block
2015	Ward member	Graduate+	-0.1	[-1.6, 1.3]	14,121	District
2015	Ward member	Age	-4.2	[-5.2, -3.2]	15,724	District
2015	District member	Graduate+	-7.8	[-21.1, 5.4]	295	District
2015	District member	Age	-3.0	[-5.8, -0.3]	327	District
2020	Block member	Graduate+	-3.5	[-7.5, 0.4]	1,850	Block
2020	Block member	Age	-3.1	[-4.0, -2.2]	2,070	Block
2020	Ward member	Graduate+	2.0	[-0.8, 4.7]	13,457	District
2020	Ward member	Age	-3.5	[-4.5, -2.4]	15,860	District
2020	District member	Graduate+	-1.8	[-15.6, 12.1]	282	District
2020	District member	Age	-3.6	[-6.0, -1.2]	331	District

References

Afridi, Farzana, Vegard Iversen, and M. R. Sharan. 2013. *Women Political Leaders, Corruption and Learning: Evidence from a Large Public Program in India*. No. 7212. IZA. <https://docs.iza.org/dp7212.pdf>.

- Bamezai, Apurva, Siddharth George, M. R. Sharan, and Borui Sun. 2026. “The Determinants and Consequences of Political Selection in the Developing World: Evidence from India.” <https://www.mrsharan.com/papers/political-selection.pdf>.
- Ban, Radu, and Vijayendra Rao. 2007. “Tokenism or Agency? The Impact of Women’s Reservations on Village Democracies in South India.” https://vijayendraraao.org/papers/EDCC_Reservations_to_go.pdf.
- Beaman, Lori, Raghavendra Chattopadhyay, Esther Duflo, Rohini Pande, and Petia Topalova. 2009. “Powerful Women: Does Exposure Reduce Bias?” *Quarterly Journal of Economics* 124 (4): 1497–540. <https://www.povertyactionlab.org/sites/default/files/research-paper/310%20Female%20Quotas%20Nov%2009.pdf>.
- Chattopadhyay, Raghavendra, and Esther Duflo. 2001. *Women as Policy Makers: Evidence from a India-Wide Randomized Policy Experiment*. No. 8615. National Bureau of Economic Research. https://www.nber.org/system/files/working_papers/w8615/w8615.pdf.
- Deininger, Klaus, Songqing Jin, Hari K. Nagarajan, and Fang Xia. 2011. *Does Female Reservation Affect Long-Term Political Outcomes? Evidence from Rural India*. No. 5708. World Bank. <https://documents1.worldbank.org/curated/en/595231468040603055/pdf/WPS5708.pdf>.
- Karekurve-Ramachandra, Varun, and Alexander Lee. 2023. *Can Gender Quotas Improve Public Service Provision? Evidence from Indian Local Government*. <https://doi.org/10.7910/DVN/IO9SLQ>.
- Mori, Yuko, R. Manjula, D. Rajasekhar, Jun Goto, and Takashi Kurosaki. 2022. “Who Becomes a Politician in a Gender Quota System? Evidence from India.” https://www.isid.ac.in/~epu/acegd2022/papers/Yuko_Mori.pdf.
- Mori, Yuko, R. Manjula, D. Rajasekhar, Jun Goto, and Takashi Kurosaki. 2025. “Who Becomes a Politician in a Gender Quota System? Evidence from India.” *World Development* 188: 106911. <https://doi.org/10.1016/j.worlddev.2024.106911>.
- Raabe, Katharina, Madhushree Sekher, and Regina Birner. 2009. *The Effects of Political Reservations for Women on Local Governance and Rural Service Provision: Survey Evidence from Karnataka*. No. 00878. International Food Policy Research Institute. <https://cgspace.cgiar.org/server/api/core/bitstreams/c85053d0-fd33-400a-b7bb-05a04874175f/content>.